KRE Prospective Evaluation: Combined Pre-Registration Package

Draft for Will's review. Not registered, not scheduled. Information cutoff: 10/30/2026 close.

What this document contains. Every file from `outputs/kre_prereg/`, combined in this order:

1. **Part 1, Pre-registration** (PREREGISTRATION.md): the claims, base rates, proposed probabilities, resolution rules and interpretation table. *This is the part to review.*
2. **Part 2, Resolver** (resolver.py): the frozen script that scores the claims after 02/02/2027. SHA-256 `b98e22f29d3f49a7bd9add75021ba4621d139e49d4d1ba707861eaf45dc9fae3`.
3. **Part 3, Base-rate scripts** (prereg_base.py, prereg_base2.py): the code that produced the base-rate tables in Part 1.
4. **Part 4, Base-rate data** (base_rate_state_series_thru_20261030.csv): the trailing 63-session correlation, volatility, beta and rate beta for every day from 2006 to 10/30/2026.

Two notes.

- In the PDF, Part 4 shows month-end values for 2006–2026 plus every day from June 2026. The full 5,059-row daily table is too long for a readable PDF; the companion `.md` version has every row.
- The code in Parts 2 and 3 is reproduced exactly, so the SHA-256 hash above still matches the standalone resolver.py.

Part 1: Pre-registration

Status: proposal only. Not registered, not scheduled, no agents contacted, no repository changes. Operational use requires Will's approval. **Information cutoff:** 10/30/2026 close. **Drafted:** 11/02/2026 (Nisaba, session 14). **Frozen resolver:** `resolver.py`, sha256 `b98e22f29d3f49a7bd9add75021ba4621d139e49d4d1ba707861eaf45dc9fae3`. It reproduces the reference-window values exactly (corr 0.7193, vol 0.1561, beta 0.3275, rate beta −0.0620).

0. Leakage disclosure (read first)

- All base rates and thresholds below use data through 10/30 only.
- **Exception:** in the earlier study-6 data pull (before this brief existed), the drafter saw near-close prices for **11/02/2026, session 1 of the evaluation window**: KRE 73.73 (+2.4%), KBWB 89.60 (+1.3%).
- The thresholds come from the brief and from earlier studies, so they were not affected. The proposed probabilities could in principle have been nudged by one of the 63 days.
- **Rule for the approver:** if you have viewed post-10/30 KRE/KBWB/SPY data before approving, record that next to the approval and mark the registration "late". It can still be scored.

1. Window

- **Evaluation window E:** the first 63 completed NYSE sessions after the 10/30/2026 close.
  - Scheduled dates: **Mon 11/02/2026 → Tue 02/02/2027.**
  - Session 21 = 12/01/2026, session 42 = 12/31/2026.
  - Full closures excluded: 11/26, 12/25, 1/1, 1/18 (NYSE calendar). The early closes on 11/27 and 12/24 count as sessions.
- **Unscheduled closures:** the window becomes the first 63 *actual* sessions, and the end date moves later.
- **Resolution date:** on or after **Wed 02/03/2027**. Target completion by 02/05/2027.
- E does not overlap the reference window (8/4–10/30). E is one independent observation, so the result is n = 1.

2. Data definitions (fixed)

1. **Returns:** daily simple returns from adjusted closes (dividends and splits) for KRE, KBWB, SPY and IWM, with base = the 10/30 close. Exactly 63 returns.
   - Primary source: Yahoo Finance v8 `adjclose`.
   - Fallback source: exchange closing prices plus issuer-published distributions.
2. **Correlation:** Pearson correlation of the 63 KRE and KBWB daily returns.
3. **Volatility:** sample standard deviation (ddof = 1) of KRE daily returns × √252.
4. **Market beta:** cov(KRE, SPY) / var(SPY) on daily returns, univariate.
5. **Rate beta:** OLS fit of r_KRE = a + b_m·r_SPY + b_rate·Δy10.
   - Δy10 = daily change in the U.S. Treasury Daily Par Yield Curve 10-year rate, in percentage points.
   - On bond-market holidays when stocks trade (e.g. Veterans Day, 11/11/2026), carry the prior yield forward.
6. **Missing data:**
   - If a close is missing from both sources, drop that session for all series. The next session's return then spans both days.
   - More than 3 dropped sessions → **VOID_DATA.**
   - If KRE or KBWB is delisted, merged, or switches benchmark index → **VOID.**
7. **Comparisons** use full precision and the inequalities exactly as written.
   - Values within the near-threshold band (corr ±0.005, vol ±0.002, beta ±0.02, rate beta ±0.005) are **flagged**, and the fallback source is cross-checked.
   - The resolution still uses the primary source.

3. Primary claims (3), base rates, proposed probabilities

Base-rate method. For every day *t* from 2006 to 2026 whose trailing 63-session state matches the condition, take the outcome over the next non-overlapping 63 sessions.

- KBWB returns are used from its launch; before 11/2011 the KBW Bank Index (BKX) stands in (daily corr 0.995 on the overlap).
- The days overlap, so "episodes" (clusters of days) are the real sample size.
- References since 2012: corr median 0.940, 10th pct 0.863; KRE vol median 23%, below 20% on only 28% of days; beta 10th pct 0.78, median 1.10.

C1. Correlation: persistence vs calm vs stress normalization

Outcome categories over E:

- **persist:** corr ≤ 0.863
- **calm_norm:** corr > 0.863 and vol < 0.20
- **stress_norm:** corr > 0.863 and vol ≥ 0.20

Conditioning state at t	episodes	persist	calm_norm	stress_norm
All days	—	0.17	0.20	0.63
corr < 0.863 (day-weighted)	13	0.36	0.17	0.46
corr < 0.863 (episode-weighted)	13	0.22	0.22	0.56
corr < 0.80	6	0.42	0.22	0.36
corr < 0.863 and vol < 0.20	9	0.12	0.42	0.46
Current episode only (Jun–Aug 2026 start dates)	1	1.00	0	0
Proposed		**0.50**	**0.20**	**0.30**

Why the proposal departs from the long-run rate.

- Of 13 historical episodes, the 2026 one is the only one that has not normalized so far.
- The current correlation, 0.72, is about 2.3 standard errors below the threshold.
- The drivers identified in study 6 (market-beta collapse since H1-2026, a bank-wide factor shift) were still present at the cutoff.
- Even so, long history says normalization is more likely than not, which is why persist is held at 0.50 rather than higher.

C2. Market-beta recovery: KRE beta to SPY over E ≥ 0.80

State	episodes	P(beta ≥ 0.80)
All days	—	0.91
beta < 0.5	3 (2018, 2024, 2026)	0.32 day-weighted / 0.50 episode-weighted
beta < 0.6	5	0.39 / 0.64
corr < 0.863 and beta < 0.6	3	0.20 / 0.41
Proposed		**0.35**

C3. Rate regime: KRE rate beta over E

Outcome categories: **neg** < −0.03 · **zero** in [−0.03, +0.03] · **pos** > +0.03. (Current value −0.062, standard error 0.023, so the band is about ±1.3 SE and classification is noisy.)

State	episodes	neg	zero	pos
All days	—	0.11	0.32	0.57
rate beta < −0.03, 2006–2026	13	0.25	0.47	0.28
rate beta < −0.03, since 2012	6	0.39	0.27	0.34
Closest analog (9/2023–7/2024)	1	0.54	—	—
Proposed		**0.40**	**0.35**	**0.25**

Scoring.

- Multi-category Brier score for C1 and C3, binary Brier for C2.
- Each is compared against the **day-weighted conditional base rate**, which is the stated reference forecast:
  - C1: 0.36 / 0.17 / 0.46
  - C2: 0.32
  - C3: 0.25 / 0.47 / 0.28
- One window cannot validate a forecaster. The scores are recorded for the ledger only.

4. Descriptive only (no prediction, no score)

- **Relative performance:** KRE − KBWB and KRE − IWM total return over E, reported separately from C1–C3. Historical 63-day KRE − KBWB standard deviation is about 6.9pp (KRE ahead 47% of the time). Study 6 showed that a gap of this size is noise-level, so it is **not** evidence for any hypothesis.
- **Optional interim readings** at sessions 21 and 42 (12/01, 12/31). These are descriptive only and **never** an early resolution.

- **Optional diagnostics** (from study 6): KRE correlation with the EW basket of traditional large banks; bank-stock beta to HY OAS changes.

5. How outcomes map to the competing interpretations

Outcome over E	"Temporary shock" (large-bank/yield shocks)	"Regime" (changed pricing of regional-bank risk)
calm_norm **and** C2 = YES	**Supported**	Weakened
calm_norm, C2 = NO	Partly supported	Unresolved (correlation back, beta still low)
stress_norm	**Unresolved.** Correlations rise mechanically when volatility spikes. Leans "temporary" only if C2 = YES as well	Not refuted
persist, vol < 0.20, C2 = NO	Weakened	**Supported**
persist, vol ≥ 0.20	Weakened	Supported (decoupling survives stress), but a different regime from the calm one

- **C3** reads separately. "neg" means the **bank-wide** inverse-yield regime is still active; that bears on the rate-relief channel, not on decoupling itself.
- C1 = persist together with C3 = zero/pos would further weaken the idea that the decoupling is rate-driven (Hypothesis A).

6. Freeze rules

- Thresholds, definitions and categories are fixed now.
- Probabilities may be changed only by Will at approval, and only if he has not viewed post-10/30 data. Otherwise the drafted values stand.
- No edits after approval. Any departure from protocol is logged, not silently applied.
- The resolver is run unchanged (hash above). If a bug is found, the fix and its effect are logged and both outputs are reported.

7. Known weaknesses

- The thresholds were chosen in-sample, on the same series that motivated the study.
- Base rates rest on 13 overlapping episodes at most. The current regime has 1–2 episodes, which matter most and are the weakest evidence.
- Yahoo `adjclose` can be revised after a dividend. Resolution uses the snapshot taken at resolution, and the data hash is recorded.
- n = 1: the outcome can support or weaken an interpretation but cannot establish it.

Part 2: Resolver (resolver.py)

How to run it, from the file's header: `--validate` reproduces the reference-window values from frozen local data; `--resolve` is for the operator only, on or after 02/03/2027, after approval. Validation output on the reference window (8/4–10/30/2026): corr 0.7193, vol 0.1561, beta 0.3275, rate beta −0.0620, C1 = persist, C2 = NO, C3 = neg.

```
"""KRE prospective evaluation: frozen resolver (draft for operator review; NOT scheduled).

Usage
  Validation on the reference window, using a local frozen CSV (no network):
      python resolver.py --validate --prices prices_adj.csv --ust-dir <dir with ust_YYYY.csv>
  Resolution (operator only, on or after 2027-02-03, after approval):
      python resolver.py --resolve
Definitions follow PREREGISTRATION.md section 2. Do not edit thresholds.
"""
import argparse, glob, io, json, hashlib, sys, time
import numpy as np, pandas as pd

TICKERS = ["KRE", "KBWB", "SPY", "IWM"]
BASE_CLOSE = "2026-10-30"
N = 63
# NYSE full closures inside the evaluation window (NYSE Group calendar). Early closes 11/27 and 12/24 ARE sessions.
NYSE_HOLIDAYS = ["2026-11-26", "2026-12-25", "2027-01-01", "2027-01-18", "2027-02-15"]
TH = dict(corr=0.863, vol=0.20, beta=0.80, brate_lo=-0.03, brate_hi=0.03)
NEAR = dict(corr=0.005, vol=0.002, beta=0.02, brate=0.005)  # flag only; does not change resolution


def expected_sessions(start="2026-11-02", n=N):
    d = pd.bdate_range(start, "2027-06-30")
    d = d[~d.isin(pd.to_datetime(NYSE_HOLIDAYS))]
    return d[:n]


def yahoo(t):
    import requests
    u = (f"https://query1.finance.yahoo.com/v8/finance/chart/{t}?period1=1761609600"
         f"&period2={int(time.time()) + 86400}&interval=1d&events=div,splits")
    j = requests.get(u, headers={"User-Agent": "Mozilla/5.0"}, timeout=30).json()["chart"]["result"][0]
    ts = pd.to_datetime(j["timestamp"], unit="s").normalize()
    return pd.Series(j["indicators"]["adjclose"][0]["adjclose"], index=ts, name=t)


def treasury_10y(years):
    import requests
    out = []
    for y in years:
        u = ("https://home.treasury.gov/resource-center/data-chart-center/interest-rates/daily-treasury-rates.csv/"
             f"{y}/all?type=daily_treasury_yield_curve&field_tdr_date_value={y}&page&_format=csv")
        x = pd.read_csv(io.StringIO(requests.get(u, headers={"User-Agent": "Mozilla/5.0"}, timeout=30).text))
        x["Date"] = pd.to_datetime(x["Date"]); out.append(x.set_index("Date")["10 Yr"].astype(float))
    s = pd.concat(out).sort_index(); return s[~s.index.duplicated()]


def stats(px, y10, sessions, base):
    """px: adjusted closes (index=dates) incl. base date; sessions: the N evaluation dates."""
    idx = pd.DatetimeIndex([pd.Timestamp(base)]).append(sessions)
    p = px.reindex(idx)
    missing = {t: [str(d.date()) for d in sessions if pd.isna(p.loc[d, t])] for t in px.columns}
    p = p.ffill()          # a missing close -> next available return spans both days (spec 2.6)
    r = p.pct_change().iloc[1:]
    for t, m in missing.items():   # make the spanned day's return zero on the missing day itself
        for d in m: r.loc[pd.Timestamp(d), t] = np.nan
    y = y10.reindex(idx).ffill()   # bond holidays (e.g. Veterans Day) carry the prior yield
    dy = y.diff().iloc[1:]
    df = pd.concat([r, dy.rename("dy")], axis=1).dropna()
    k, b, s = df.KRE, df.KBWB, df.SPY
    corr = float(np.corrcoef(k, b)[0, 1])
    vol = float(k.std(ddof=1) * np.sqrt(252))
    beta = float(np.cov(k, s, ddof=1)[0, 1] / s.var(ddof=1))
    X = np.column_stack([np.ones(len(df)), s, df.dy])
    brate = float(np.linalg.lstsq(X, k, rcond=None)[0][2])
    cum = lambda t: float((1 + r[t].fillna(0)).prod() - 1)
    out = dict(n_returns_used=int(len(df)), missing=missing, corr=corr, vol=vol, beta=beta, brate=brate,
               KRE_ret=cum("KRE"), KBWB_ret=cum("KBWB"), IWM_ret=cum("IWM") if "IWM" in r else None)
    out["KRE_minus_KBWB"] = out["KRE_ret"] - out["KBWB_ret"]
    if out["IWM_ret"] is not None: out["KRE_minus_IWM"] = out["KRE_ret"] - out["IWM_ret"]
    out["C1"] = "persist" if corr <= TH["corr"] else ("calm_norm" if vol < TH["vol"] else "stress_norm")
    out["C2"] = "YES" if beta >= TH["beta"] else "NO"
    out["C3"] = "neg" if brate < TH["brate_lo"] else ("pos" if brate > TH["brate_hi"] else "zero")
    out["near_threshold"] = {
        "corr": abs(corr - TH["corr"]) < NEAR["corr"], "vol": abs(vol - TH["vol"]) < NEAR["vol"],
        "beta": abs(beta - TH["beta"]) < NEAR["beta"],
        "brate": min(abs(brate - TH["brate_lo"]), abs(brate - TH["brate_hi"])) < NEAR["brate"]}
    nmiss = max(len(v) for v in missing.values())
    out["status"] = "VOID_DATA" if nmiss > 3 else "RESOLVED"
    return out


def main():
    ap = argparse.ArgumentParser()
    ap.add_argument("--validate", action="store_true"); ap.add_argument("--resolve", action="store_true")
```

```
    ap.add_argument("--prices"); ap.add_argument("--ust-dir")
    a = ap.parse_args()
    if a.validate:   # reference window 8/4-10/30/2026 from frozen local files; expected 0.7193/0.1561/0.3275/-0.062
        px = pd.read_csv(a.prices, index_col=0, parse_dates=True)[TICKERS].loc[:"2026-10-30"]
        px = px[px.SPY.notna()]
        u = [pd.read_csv(f) for f in sorted(glob.glob(a.ust_dir + "/ust_*.csv"))]
        for x in u: x["Date"] = pd.to_datetime(x["Date"])
        y10 = pd.concat([x.set_index("Date")["10 Yr"] for x in u]).astype(float).sort_index()
        y10 = y10[~y10.index.duplicated()]
        sess = px.loc["2026-08-04":"2026-10-30"].index
        print(json.dumps(stats(px, y10, sess, "2026-08-03"), indent=1)); return
    if a.resolve:
        sess = expected_sessions()
        if pd.Timestamp.today().normalize() <= sess[-1]:
            sys.exit(f"Not resolvable before session 63 ({sess[-1].date()}) has closed.")
        px = pd.concat([yahoo(t) for t in TICKERS], axis=1)
        traded = px.index[(px.index > BASE_CLOSE)]
        # unscheduled closures: the window is the first 63 ACTUAL sessions (spec 2.1)
        actual = traded[:N]
        if not actual.equals(sess): print("NOTE: actual sessions differ from scheduled calendar; using actual.")
        y10 = treasury_10y([2026, 2027])
        res = stats(px, y10, actual, BASE_CLOSE)
        res["window"] = [str(actual[0].date()), str(actual[-1].date())]
        res["data_sha256"] = hashlib.sha256(px.to_csv().encode()).hexdigest()
        res["retrieved_utc"] = pd.Timestamp.utcnow().isoformat()
        print(json.dumps(res, indent=1))


if __name__ == "__main__":
    main()
```

Part 3: Base-rate scripts

prereg_base.py: builds the daily state series and conditional base rates

The KBW Bank Index (BKX) stands in for KBWB before 11/2011, Treasury par 10-year yields cover 2005–2026, and all data are truncated at 10/30/2026.

```
        import pandas as pd, numpy as np, glob
        D='/tmp/s6/data/'
        adj=pd.read_csv(D+'prices_adj.csv',index_col=0,parse_dates=True).sort_index()
        adj=adj[adj['SPY'].notna()].loc[:'2026-10-30']          # FREEZE: nothing after 10/30
        u=[]
        for f in sorted(glob.glob(D+'ust_*.csv')):
            x=pd.read_csv(f); x['Date']=pd.to_datetime(x['Date']); u.append(x.set_index('Date'))
        ust=pd.concat(u).sort_index(); ust=ust[~ust.index.duplicated()]
        y10=ust['10 Yr'].astype(float).reindex(adj.index).ffill()
        r=adj.pct_change()
        # large-bank series: KBWB where available, else ^BKX (price index) before KBWB inception
        lb=r['KBWB'].copy(); bkx=r['^BKX']
        print('KBWB vs ^BKX daily corr (overlap):',round(lb.corr(bkx),4))
        lb=lb.where(lb.notna(),bkx)
        N=63
        df=pd.DataFrame({'KRE':r.KRE,'LB':lb,'SPY':r.SPY,'dy':y10.diff()}).loc['2006-06-23':].dropna()
        corr=df.KRE.rolling(N).corr(df.LB)
        vol=df.KRE.rolling(N).std()*np.sqrt(252)
        beta=df.KRE.rolling(N).cov(df.SPY)/df.SPY.rolling(N).var()
        # rate beta: OLS KRE ~ SPY + dy (pp)
        def rb(i):
            s=df.iloc[i-N+1:i+1]; X=np.column_stack([np.ones(N),s.SPY,s.dy]); return np.linalg.lstsq(X,s.KRE,rcond=None)[0][2]
        brate=pd.Series([rb(i) if i>=N-1 else np.nan for i in range(len(df))],index=df.index)
        S=pd.DataFrame({'corr':corr,'vol':vol,'beta':beta,'brate':brate}).dropna()
        cur=S.iloc[-1]; print('CURRENT (10/30):',cur.round(4).to_dict())
        # histories for reference thresholds (since 2012-02 like study 1, KBWB era)
        h=S.loc['2012-02-01':]
        print('since 2012-02: corr median %.3f p10 %.3f | vol median %.3f | beta median %.2f p10 %.2f | brate median %.3f'%
(h['corr'].median(),h['corr'].quantile(.1),h.vol.median(),h.beta.median(),h.beta.quantile(.1),h.brate.median()))
        # forward (non-overlapping next 63) values
        F=S.shift(-N).add_suffix('_f')
        J=pd.concat([S,F],axis=1).dropna()
        J['f_corr_ok']=J.corr_f>0.863; J['f_calm']=J.vol_f<0.20; J['f_P1']=J.f_corr_ok&J.f_calm
        J['f_P2']=J.beta_f>=0.80; J['f_P3']=J.brate_f< -0.03
        def episodes(mask):
            idx=J.index[mask];
            if len(idx)==0: return 0
            g=(pd.Series(idx).diff()>pd.Timedelta(days=95)).cumsum(); return g.nunique()
        def report(name,mask):
            sub=J[mask]
            if len(sub)==0: print(name,'n=0'); return
            print(f'{name:58s} days={len(sub):5d} episodes={episodes(mask):2d} | P1 {sub.f_P1.mean():.2f} corr>.863 {sub.f_corr_ok.mean():.2f}
beta>=.8 {sub.f_P2.mean():.2f} | P3 brate<-.03 {sub.f_P3.mean():.2f} | fwd corr med {sub.corr_f.median():.3f}')
        print('--- conditional base rates (state at t -> next non-overlapping 63 sessions) ---')
        report('ALL days 2006-2026',J.index==J.index)
        report('ALL days since 2012-02',J.index>=pd.Timestamp('2012-02-01'))
        report('corr<0.863',J['corr']<0.863)
        report('corr<0.863, pre-2025',(J['corr']<0.863)&(J.index<'2025-01-01'))
        report('corr<0.863, 2025+',(J['corr']<0.863)&(J.index>='2025-01-01'))
        report('corr<0.80',J['corr']<0.80)
        report('corr<0.80 & vol<0.20',(J['corr']<0.80)&(J.vol<0.20))
        report('beta<0.5',J.beta<0.5)
        report('beta<0.5, pre-2025',(J.beta<0.5)&(J.index<'2025-01-01'))
        report('beta<0.5, 2025+',(J.beta<0.5)&(J.index>='2025-01-01'))
        report('brate<-0.03',J.brate<-0.03)
        report('brate<-0.03, since 2012',(J.brate<-0.03)&(J.index>='2012-01-01'))
        report('brate<-0.03, since 2023-08',(J.brate<-0.03)&(J.index>='2023-08-01'))
        report('brate<0, since 2012',(J.brate<0)&(J.index>='2012-01-01'))
        report('joint: corr<.863 & beta<.5 & brate<0',(J['corr']<0.863)&(J.beta<0.5)&(J.brate<0))
        # strict non-overlapping grid anchored at 10/30 (independent obs)
        g=S.iloc[::-1].iloc[::N].iloc[::-1]; G=pd.DataFrame({'corr':g['corr'],'vol':g.vol,'beta':g.beta,'brate':g.brate})
        G['corr_f']=G['corr'].shift(-1); G['vol_f']=G.vol.shift(-1); G['beta_f']=G.beta.shift(-1); G['brate_f']=G.brate.shift(-1)
        Gd=G.dropna(); print('--- non-overlapping grid (anchored 10/30), n windows with fwd:',len(Gd))
        for nm,m in [('all',Gd.index==Gd.index),('corr<.863',Gd['corr']<.863),('beta<.5',Gd.beta<.5),('brate<-.03',Gd.brate<-.03)]:
            s=Gd[m]; print(f'  {nm:12s} n={len(s):3d} P1 {((s.corr_f>.863)&(s.vol_f<.2)).mean():.2f} corr>.863 {(s.corr_f>.863).mean():.2f} P2
{(s.beta_f>=.8).mean():.2f} P3 {(s.brate_f<-.03).mean():.2f}')
        print(Gd[Gd['corr']<.863][['corr','vol','beta','corr_f','vol_f','beta_f','brate','brate_f']].round(3).to_string())
        print('recent grid windows:'); print(G.tail(8).round(3).to_string())
        # rate-beta SE check (current window)
        import statsmodels.api as sm
        s=df.iloc[-N:]; m=sm.OLS(s.KRE,sm.add_constant(s[['SPY','dy']])).fit(); print('current brate %.3f se %.3f'%(m.params['dy'],m.bse['dy']
        # relative performance distribution (descriptive)
        lbc=(1+df.LB).rolling(N).apply(np.prod,raw=True)-1; krc=(1+df.KRE).rolling(N).apply(np.prod,raw=True)-1
        iw=(1+r.IWM.reindex(df.index)).rolling(N).apply(np.prod,raw=True)-1
        gap=(krc-lbc).dropna(); gi=(krc-iw).dropna()
        print('63d KRE-LB gap: share>0 %.2f, sd %.3f | KRE-IWM share>0 %.2f sd %.3f'%((gap>0).mean(),gap.std(),(gi>0).mean(),gi.std()))
        S.to_csv('/tmp/s6/out/prereg_state_series.csv')
```

prereg_base2.py: three-way outcome distributions and episode lists

```
        import pandas as pd, numpy as np
        S=pd.read_csv('/tmp/s6/out/prereg_state_series.csv',index_col=0,parse_dates=True)
```

```
        N=63; F=S.shift(-N).add_suffix('_f'); J=pd.concat([S,F],axis=1).dropna()
        def c1(r): return np.where(r.corr_f<=0.863,'persist',np.where(r.vol_f<0.20,'calm_norm','stress_norm'))
        def c3(r): return np.where(r.brate_f< -0.03,'neg',np.where(r.brate_f>0.03,'pos','zero'))
        J['C1']=c1(J); J['C3']=c3(J); J['C2']=np.where(J.beta_f>=0.8,'beta>=.8','beta<.8')
        def eps(mask,gap=63):
            idx=J.index[mask];
            if len(idx)==0: return []
            s=pd.Series(idx); g=(s.diff()>pd.Timedelta(days=int(gap*1.5))).cumsum()
            return [(grp.iloc[0].date(),grp.iloc[-1].date(),len(grp)) for _,grp in s.groupby(g)]
        sets={'all':J.index==J.index,'corr<.863':J['corr']<.863,'corr<.80':J['corr']<.80,'corr<.863 & vol<.20':(J['corr']<.863)&(J.vol<.20),
              'beta<.5':J.beta<.5,'beta<.6':J.beta<.6,'brate<-.03':J.brate<-.03,'brate<-.03 & since2012':(J.brate<-.03)&(J.index>='2012-01-01'
              'corr<.863 & beta<.6':(J['corr']<.863)&(J.beta<.6),'vol<.20':J.vol<.20}
        for nm,m in sets.items():
            s=J[m]; print(f'\n## {nm}: days={len(s)} episodes={len(eps(m))}')
            print('  C1',s.C1.value_counts(normalize=True).round(2).to_dict(),'| C2',s.C2.value_counts(normalize=True).round(2).to_dict(),'|
C3',s.C3.value_counts(normalize=True).round(2).to_dict())
            # episode-level (each episode weighted equally)
            E=[]
            for a,b,n in eps(m):
                e=J.loc[str(a):str(b)][m.loc[str(a):str(b)] if hasattr(m,'loc') else slice(None)]
                E.append({'start':a,'end':b,'days':n,'corr>.863':(e.corr_f>.863).mean(),'calm_norm':(e.C1=='calm_norm').mean(),'beta>=.8':
(e.beta_f>=.8).mean(),'brate_neg':(e.C3=='neg').mean()})
            if nm!='all' and nm!='vol<.20':
                E=pd.DataFrame(E); print(E.round(2).to_string(index=False)); print('  episode-
weighted:',E[['corr>.863','calm_norm','beta>=.8','brate_neg']].mean().round(2).to_dict())
        # persistence of vol<.20
        print('\nP(vol_f<.20 | vol<.20)=%.2f ; P(vol_f<.20) all=%.2f'%((J[J.vol<.2].vol_f<.2).mean(),(J.vol_f<.2).mean()))
        print('share of days since 2006 with vol<.20: %.2f'%((S.vol<.2).mean()))
        print('se of 63d corr near 0.72 (Fisher): %.3f'%((1-0.72**2)/np.sqrt(60)))
```

Part 4: Base-rate data (state series through 10/30/2026)

Columns: trailing 63-session KRE–KBWB correlation; KRE annualized volatility; KRE beta to SPY; KRE rate beta (from KRE regressed on SPY and Δ10y, per percentage point). Each row describes the 63 sessions ending on that date. Data begin in 2006, once 63 sessions of history are available.

Month-end values, 2006–2026

Date	corr	vol	beta	rate beta
2006-09-29	0.8895	0.1346	1.1695	-0.0299
2006-10-31	0.7590	0.1136	1.2459	-0.0044
2006-11-30	0.7453	0.1178	1.2165	0.0116
2006-12-29	0.7458	0.1168	1.2536	0.0085
2007-01-31	0.7570	0.1216	1.2204	0.0056
2007-02-28	0.8548	0.1363	1.0009	-0.0120
2007-03-30	0.9056	0.1636	1.0589	-0.0133
2007-04-30	0.9310	0.1608	1.0281	-0.0149
2007-05-31	0.9070	0.1525	1.2045	-0.0194
2007-06-29	0.8619	0.1300	0.8993	-0.0109
2007-07-31	0.8899	0.1581	0.9203	0.0180
2007-08-31	0.8937	0.2854	1.1242	0.0012
2007-09-28	0.9039	0.3015	1.2484	-0.0222
2007-10-31	0.9240	0.3145	1.3585	-0.0329
2007-11-30	0.9490	0.3498	1.4575	0.0357
2007-12-31	0.9333	0.3847	1.5345	0.0522
2008-01-31	0.9492	0.4648	1.6520	-0.0009
2008-02-29	0.9370	0.4149	1.7003	-0.0135
2008-03-31	0.9316	0.4362	1.5166	-0.0441
2008-04-30	0.9040	0.4037	1.4620	-0.0072
2008-05-30	0.8996	0.3842	1.4198	0.0013
2008-06-30	0.8680	0.3334	1.1578	0.0354
2008-07-31	0.9411	0.5512	1.9215	0.0102
2008-08-29	0.9334	0.5971	2.0583	0.0226
2008-09-30	0.8133	0.7356	1.5155	-0.0645
2008-10-31	0.7610	0.8329	1.0505	-0.0212
2008-11-28	0.7956	0.9082	1.0312	-0.0031
2008-12-31	0.8916	0.9011	1.1092	0.0354
2009-01-30	0.9314	0.7958	1.3598	-0.0222
2009-02-27	0.9417	0.7332	1.6117	-0.0331
2009-03-31	0.9440	0.8134	1.7719	-0.0600
2009-04-30	0.9461	0.8333	1.9043	-0.0322
2009-05-29	0.9424	0.8549	2.1540	-0.0173
2009-06-30	0.9291	0.6608	2.2489	-0.0103
2009-07-31	0.8952	0.5289	1.8857	-0.0255
2009-08-31	0.8321	0.3951	1.5455	0.0206
2009-09-30	0.8269	0.3903	1.6142	0.0341
2009-10-30	0.7777	0.3657	1.6345	0.0244
2009-11-30	0.8040	0.3312	1.4639	-0.0302
2009-12-31	0.7899	0.2796	1.2754	-0.0122
2010-01-29	0.7544	0.2845	1.0733	-0.0180
2010-02-26	0.7264	0.2525	0.8299	-0.0061
2010-03-31	0.6935	0.2303	0.7744	-0.0172
2010-04-30	0.8622	0.2345	1.0936	-0.0206
2010-05-28	0.9228	0.3202	1.3448	-0.0229
2010-06-30	0.9294	0.3646	1.3050	-0.0154
2010-07-30	0.9354	0.3853	1.3503	0.0400
2010-08-31	0.9141	0.3298	1.4220	0.0362
2010-09-30	0.9263	0.3267	1.6200	0.0193
2010-10-29	0.8849	0.2571	1.4764	-0.0108

2010-11-30	0.8785	0.2527	1.4230	-0.0141
2010-12-31	0.8222	0.2311	1.3684	-0.0058
2011-01-31	0.8146	0.2534	1.5429	0.0024
2011-02-28	0.8002	0.2266	1.6247	0.0037
2011-03-31	0.8097	0.2135	1.1751	0.0047
2011-04-29	0.8367	0.1788	1.0469	-0.0222
2011-05-31	0.8379	0.1696	1.0146	-0.0094
2011-06-30	0.8429	0.1737	1.0541	0.0054
2011-07-29	0.8817	0.1915	1.1033	0.0037
2011-08-31	0.9661	0.4210	1.3134	0.0020
2011-09-30	0.9683	0.5095	1.4177	-0.0002
2011-10-31	0.9643	0.6024	1.5191	-0.0152
2011-11-30	0.9296	0.5348	1.6248	-0.0576
2011-12-30	0.9003	0.4655	1.6011	-0.0540
2012-01-31	0.8520	0.3451	1.4815	-0.0092
2012-02-29	0.8486	0.2719	1.6110	-0.0181
2012-03-30	0.9148	0.2043	1.7317	0.0349
2012-04-30	0.9342	0.2201	1.6400	0.0545
2012-05-31	0.9337	0.2061	1.3532	0.0617
2012-06-29	0.9533	0.2241	1.2221	0.0270
2012-07-31	0.9432	0.2061	1.0915	0.0043
2012-08-31	0.9218	0.1731	1.0484	0.0228
2012-09-28	0.8868	0.1532	1.0471	0.0573
2012-10-31	0.8652	0.1562	0.9773	0.0672
2012-11-30	0.9011	0.1843	1.0892	0.0720
2012-12-31	0.8970	0.1762	1.0387	0.0590
2013-01-31	0.9015	0.1737	1.2263	0.0408
2013-02-28	0.8861	0.1430	1.0690	0.0375
2013-03-28	0.8952	0.1327	1.0090	0.0411
2013-04-30	0.9188	0.1450	1.0708	0.0214
2013-05-31	0.8770	0.1518	1.1591	0.0149
2013-06-28	0.8862	0.1740	1.0096	0.0418
2013-07-31	0.8517	0.1623	0.8546	0.0540
2013-08-30	0.9208	0.1809	0.9697	0.0635
2013-09-30	0.8772	0.1603	0.8760	0.0529
2013-10-31	0.8675	0.1539	0.9771	0.0290
2013-11-29	0.8513	0.1514	0.9488	0.0351
2013-12-31	0.8743	0.1602	1.2112	0.0392
2014-01-31	0.8703	0.1728	1.2810	0.0341
2014-02-28	0.8674	0.1852	1.2618	0.0222
2014-03-31	0.8933	0.1896	1.2473	0.0429
2014-04-30	0.8915	0.1940	1.2188	0.0492
2014-05-30	0.8726	0.1714	1.1532	0.0828
2014-06-30	0.8630	0.1636	1.2005	0.0558
2014-07-31	0.9141	0.1617	1.2794	0.0696
2014-08-29	0.8960	0.1454	1.0624	0.0681
2014-09-30	0.8960	0.1445	0.9208	0.1042
2014-10-31	0.8636	0.1713	0.9163	0.0866
2014-11-28	0.8382	0.1779	0.9294	0.0799
2014-12-31	0.8524	0.1998	1.0054	0.0532
2015-01-30	0.9153	0.2277	1.3318	0.0493
2015-02-27	0.9387	0.2454	1.4058	0.0594
2015-03-31	0.9545	0.2311	1.2524	0.0859
2015-04-30	0.9274	0.1704	0.9389	0.0716
2015-05-29	0.8934	0.1442	0.6193	0.0737
2015-06-30	0.8986	0.1451	0.7199	0.0829
2015-07-31	0.8962	0.1550	0.7483	0.0892
2015-08-31	0.9331	0.2092	0.8915	0.0620
2015-09-30	0.9415	0.2276	0.8800	0.0589
2015-10-30	0.9430	0.2653	0.9680	0.1047

2015-11-30	0.9381	0.2347	1.0228	0.1639
2015-12-31	0.9352	0.2329	1.1710	0.0936
2016-01-29	0.9517	0.2437	1.0823	0.0398
2016-02-29	0.9731	0.2880	1.2100	0.0607
2016-03-31	0.9779	0.3001	1.3352	0.1019
2016-04-29	0.9822	0.2765	1.6678	0.1060
2016-05-31	0.9713	0.2244	1.6769	0.1221
2016-06-30	0.9883	0.3055	1.9162	0.1653
2016-07-29	0.9849	0.2941	1.8944	0.1643
2016-08-31	0.9857	0.2751	1.9659	0.1168
2016-09-30	0.9573	0.1715	1.0948	0.1200
2016-10-31	0.9569	0.1669	1.0169	0.1418
2016-11-30	0.9643	0.2290	1.1413	0.2098
2016-12-30	0.9659	0.2245	1.3665	0.1737
2017-01-31	0.9666	0.2390	1.5639	0.1733
2017-02-28	0.9608	0.1904	1.7192	0.1277
2017-03-31	0.9647	0.2245	2.4511	0.1123
2017-04-28	0.9511	0.2213	2.4228	0.0857
2017-05-31	0.9521	0.2174	2.2471	0.1083
2017-06-30	0.9434	0.1985	1.5213	0.1472
2017-07-31	0.9433	0.1800	1.0676	0.1955
2017-08-31	0.9403	0.1725	0.9682	0.1716
2017-09-29	0.9574	0.1867	1.6398	0.2010
2017-10-31	0.9564	0.1864	1.8385	0.1821
2017-11-30	0.9478	0.1957	1.5519	0.1573
2017-12-29	0.9294	0.1763	1.3273	0.0594
2018-01-31	0.9245	0.1706	0.9652	0.0247
2018-02-28	0.9457	0.2151	0.9086	0.0933
2018-03-29	0.9513	0.2271	0.9599	0.0945
2018-04-30	0.9604	0.2463	0.9209	0.1344
2018-05-31	0.9507	0.2103	1.0064	0.1366
2018-06-29	0.9179	0.1616	0.8140	0.1292
2018-07-31	0.8614	0.1424	0.7511	0.1282
2018-08-31	0.8187	0.1333	0.6007	0.1109
2018-09-28	0.8365	0.1423	0.5632	0.1220
2018-10-31	0.8981	0.1992	0.7331	0.1186
2018-11-30	0.9218	0.2101	0.6926	0.1075
2018-12-31	0.9341	0.2682	0.7864	0.1245
2019-01-31	0.9464	0.2444	0.7786	0.0935
2019-02-28	0.9391	0.2429	0.7978	0.0958
2019-03-29	0.9348	0.2094	0.7863	0.1278
2019-04-30	0.9416	0.2158	1.2106	0.1509
2019-05-31	0.9604	0.2411	1.4175	0.1584
2019-06-28	0.9575	0.2138	1.2742	0.1004
2019-07-31	0.9530	0.2059	1.0876	0.0966
2019-08-30	0.9619	0.2364	1.1381	0.1181
2019-09-30	0.9663	0.2511	1.2046	0.1587
2019-10-31	0.9745	0.2405	1.1752	0.1436
2019-11-29	0.9683	0.1913	1.2780	0.1330
2019-12-31	0.9590	0.1571	1.2094	0.1015
2020-01-31	0.9576	0.1568	1.2848	0.0989
2020-02-28	0.9764	0.2107	1.0644	0.1083
2020-03-31	0.9829	0.7496	1.2906	0.0534
2020-04-30	0.9830	0.8840	1.3292	0.0600
2020-05-29	0.9821	0.9653	1.4190	0.0568
2020-06-30	0.9834	0.7341	1.8596	0.1668
2020-07-31	0.9760	0.6265	2.1577	0.1859
2020-08-31	0.9716	0.5114	1.6734	0.3954
2020-09-30	0.9632	0.3750	0.6562	0.3407
2020-10-30	0.9676	0.3436	0.7415	0.3836

2020-11-30	0.9816	0.5016	1.0036	0.5264
2020-12-31	0.9801	0.4745	1.2202	0.5088
2021-01-29	0.9796	0.4807	1.3410	0.4785
2021-02-26	0.9620	0.3108	1.2684	0.1898
2021-03-31	0.9643	0.3372	0.9881	0.1938
2021-04-30	0.9592	0.3001	1.0206	0.1747
2021-05-28	0.9482	0.2734	0.8581	0.1794
2021-06-30	0.9436	0.2692	1.2486	0.2288
2021-07-30	0.9511	0.2890	1.4549	0.2028
2021-08-31	0.9529	0.2968	1.9055	0.2098
2021-09-30	0.9533	0.2798	1.6028	0.1686
2021-10-29	0.9319	0.2398	1.1913	0.1310
2021-11-30	0.9258	0.2475	1.1792	0.1177
2021-12-31	0.9205	0.2457	1.1854	0.1219
2022-01-31	0.9289	0.2685	1.0234	0.1275
2022-02-28	0.9405	0.2725	0.8730	0.0935
2022-03-31	0.9464	0.3280	0.9755	0.1157
2022-04-29	0.9465	0.3249	0.9728	0.0887
2022-05-31	0.9506	0.3110	0.9234	0.0584
2022-06-30	0.9500	0.2812	0.8590	0.0134
2022-07-29	0.9549	0.2807	0.8701	0.0155
2022-08-31	0.9456	0.2484	0.9071	0.0154
2022-09-30	0.9412	0.2241	0.8608	-0.0078
2022-10-31	0.9457	0.2634	0.9247	-0.0191
2022-11-30	0.9427	0.2790	0.8966	-0.0123
2022-12-30	0.9385	0.2785	0.9400	0.0057
2023-01-31	0.9232	0.2510	0.9655	0.0239
2023-02-28	0.9105	0.2330	1.0468	0.0218
2023-03-31	0.9595	0.4375	1.7438	0.1139
2023-04-28	0.9420	0.4498	1.8177	0.1364
2023-05-31	0.9477	0.5254	2.4285	0.1137
2023-06-30	0.9397	0.4217	2.1154	0.0789
2023-07-31	0.9651	0.4291	2.3247	0.0343
2023-08-31	0.9507	0.3255	1.2496	-0.0016
2023-09-29	0.9339	0.2655	1.1889	-0.0427
2023-10-31	0.9304	0.2427	0.9659	-0.0023
2023-11-30	0.9432	0.3110	1.5212	-0.0293
2023-12-29	0.9528	0.3493	1.8004	-0.0646
2024-01-31	0.9353	0.3490	2.3176	-0.0515
2024-02-29	0.9177	0.3179	1.6342	-0.0403
2024-03-28	0.8787	0.2810	1.0671	-0.0146
2024-04-30	0.8968	0.2896	0.8269	-0.0663
2024-05-31	0.8638	0.2438	0.7109	-0.1570
2024-06-28	0.8725	0.2296	0.7539	-0.0546
2024-07-31	0.8727	0.2523	0.3978	-0.0271
2024-08-30	0.8891	0.2928	0.8344	0.0502
2024-09-30	0.8890	0.2937	0.9055	0.0422
2024-10-31	0.8930	0.2507	0.9960	0.0058
2024-11-29	0.9634	0.3585	1.5695	0.0996
2024-12-31	0.9743	0.3574	1.7105	0.0663
2025-01-31	0.9639	0.3554	1.5502	0.0510
2025-02-28	0.8905	0.2245	1.1036	-0.0634
2025-03-31	0.9104	0.2377	1.0479	-0.0341
2025-04-30	0.9592	0.3765	0.9598	0.0154
2025-05-30	0.9707	0.3902	1.0024	0.0207
2025-06-30	0.9655	0.3786	1.0044	0.0264
2025-07-31	0.9160	0.2395	1.5493	0.0074
2025-08-29	0.8858	0.2339	1.6124	-0.0266
2025-09-30	0.8464	0.2106	1.4205	-0.0290
2025-10-31	0.8784	0.2592	1.4123	0.0161

2025-11-28	0.8452	0.2536	1.0402	0.0422
2025-12-31	0.8607	0.2550	1.0256	0.0157
2026-01-30	0.8108	0.2186	0.8361	-0.0512
2026-02-27	0.8657	0.2449	0.9493	0.0220
2026-03-31	0.8835	0.2522	0.9201	0.0464
2026-04-30	0.9083	0.2329	0.8379	0.0460
2026-05-29	0.8851	0.1901	0.7989	-0.0222
2026-06-30	0.8130	0.2033	0.4180	-0.1347
2026-07-31	0.7279	0.1982	0.2761	-0.1220
2026-08-31	0.6999	0.1843	0.2247	-0.0563
2026-09-30	0.7033	0.1625	0.5725	0.0040
2026-10-30	0.7193	0.1561	0.3275	-0.0620

Every day from 06/01/2026 to 10/30/2026

Date	corr	vol	beta	rate beta
2026-06-01	0.8771	0.1919	0.7979	-0.0424
2026-06-02	0.8800	0.1946	0.7989	-0.0467
2026-06-03	0.8805	0.2006	0.8311	-0.0450
2026-06-04	0.8894	0.2078	0.8316	-0.0513
2026-06-05	0.8752	0.2008	0.6712	-0.0679
2026-06-08	0.8739	0.2007	0.6831	-0.0693
2026-06-09	0.8736	0.2018	0.6719	-0.0768
2026-06-10	0.8623	0.2002	0.6212	-0.0786
2026-06-11	0.8698	0.1998	0.6165	-0.0747
2026-06-12	0.8711	0.2006	0.6185	-0.0691
2026-06-15	0.8692	0.2040	0.5454	-0.0831
2026-06-16	0.8661	0.2039	0.5382	-0.0850
2026-06-17	0.8594	0.2059	0.5517	-0.0861
2026-06-18	0.8512	0.2057	0.5604	-0.0837
2026-06-22	0.8531	0.2060	0.5641	-0.0919
2026-06-23	0.8483	0.2054	0.4840	-0.1018
2026-06-24	0.8358	0.2058	0.4875	-0.1070
2026-06-25	0.8367	0.2065	0.4872	-0.1117
2026-06-26	0.8254	0.2064	0.4950	-0.1174
2026-06-29	0.8163	0.2032	0.4175	-0.1267
2026-06-30	0.8130	0.2033	0.4180	-0.1347
2026-07-01	0.8099	0.2009	0.3447	-0.1308
2026-07-02	0.8051	0.2033	0.3497	-0.1368
2026-07-06	0.8008	0.2036	0.3535	-0.1376
2026-07-07	0.7996	0.2038	0.3590	-0.1424
2026-07-08	0.8123	0.2099	0.3842	-0.1404
2026-07-09	0.8026	0.2051	0.3028	-0.1485
2026-07-10	0.7997	0.2028	0.2908	-0.1482
2026-07-13	0.7975	0.2007	0.2776	-0.1445
2026-07-14	0.7910	0.2005	0.2705	-0.1406
2026-07-15	0.7924	0.2012	0.2907	-0.1432
2026-07-16	0.7588	0.2080	0.2533	-0.1457
2026-07-17	0.7649	0.2106	0.2945	-0.1281
2026-07-20	0.7640	0.2079	0.2539	-0.1258
2026-07-21	0.7625	0.2078	0.2524	-0.1235
2026-07-22	0.7645	0.2058	0.2302	-0.1224
2026-07-23	0.7646	0.2059	0.2640	-0.1206
2026-07-24	0.7727	0.2054	0.2761	-0.1248
2026-07-27	0.7664	0.2026	0.3096	-0.1232
2026-07-28	0.7567	0.2025	0.3119	-0.1367
2026-07-29	0.7501	0.2033	0.3338	-0.1367
2026-07-30	0.7313	0.1993	0.2933	-0.1305
2026-07-31	0.7279	0.1982	0.2761	-0.1220
2026-08-03	0.7311	0.1995	0.2993	-0.1236

2026-08-04	0.7271	0.1983	0.2982	-0.1201
2026-08-05	0.7211	0.1976	0.2898	-0.1185
2026-08-06	0.7233	0.1983	0.2800	-0.1212
2026-08-07	0.7176	0.1970	0.2638	-0.1151
2026-08-10	0.7178	0.1971	0.2722	-0.1138
2026-08-11	0.7084	0.1935	0.2707	-0.1092
2026-08-12	0.7133	0.1933	0.2696	-0.1088
2026-08-13	0.7018	0.1898	0.2901	-0.1057
2026-08-14	0.7001	0.1894	0.2831	-0.1057
2026-08-17	0.6977	0.1883	0.2642	-0.1058
2026-08-18	0.7001	0.1877	0.2797	-0.1045
2026-08-19	0.7208	0.1943	0.2637	-0.0839
2026-08-20	0.6995	0.1894	0.2289	-0.0722
2026-08-21	0.6988	0.1894	0.2291	-0.0698
2026-08-24	0.6969	0.1895	0.2304	-0.0668
2026-08-25	0.6919	0.1884	0.2113	-0.0560
2026-08-26	0.6875	0.1871	0.2098	-0.0586
2026-08-27	0.6881	0.1872	0.2054	-0.0606
2026-08-28	0.6892	0.1873	0.2065	-0.0596
2026-08-31	0.6999	0.1843	0.2247	-0.0563
2026-09-01	0.6936	0.1833	0.2392	-0.0566
2026-09-02	0.6926	0.1812	0.2210	-0.0534
2026-09-03	0.6529	0.1717	0.2106	-0.0485
2026-09-04	0.6577	0.1719	0.2509	-0.0505
2026-09-08	0.6661	0.1742	0.2691	-0.0512
2026-09-09	0.6622	0.1744	0.2942	-0.0481
2026-09-10	0.6730	0.1743	0.3252	-0.0352
2026-09-11	0.6683	0.1733	0.3032	-0.0342
2026-09-14	0.6330	0.1709	0.2808	-0.0391
2026-09-15	0.6324	0.1677	0.3897	-0.0350
2026-09-16	0.6575	0.1715	0.4218	-0.0311
2026-09-17	0.6653	0.1672	0.3537	-0.0244
2026-09-18	0.6762	0.1665	0.3410	-0.0232
2026-09-21	0.6614	0.1658	0.3081	-0.0245
2026-09-22	0.6671	0.1642	0.4089	-0.0120
2026-09-23	0.6864	0.1639	0.4327	-0.0074
2026-09-24	0.6771	0.1631	0.4259	-0.0001
2026-09-25	0.6999	0.1638	0.4599	0.0048
2026-09-28	0.7108	0.1656	0.5476	0.0068
2026-09-29	0.7167	0.1667	0.5582	0.0087
2026-09-30	0.7033	0.1625	0.5725	0.0040
2026-10-01	0.7059	0.1609	0.5683	-0.0020
2026-10-02	0.7067	0.1621	0.5794	0.0025
2026-10-05	0.7068	0.1620	0.5579	0.0003
2026-10-06	0.6810	0.1558	0.5221	-0.0003
2026-10-07	0.6679	0.1546	0.4985	0.0023
2026-10-08	0.6706	0.1557	0.4713	-0.0098
2026-10-09	0.6603	0.1563	0.4710	-0.0137
2026-10-12	0.6699	0.1574	0.4687	-0.0159
2026-10-13	0.6629	0.1561	0.4584	-0.0106
2026-10-14	0.7138	0.1446	0.5253	-0.0079
2026-10-15	0.7168	0.1470	0.5084	-0.0268
2026-10-16	0.7136	0.1459	0.4715	-0.0275
2026-10-19	0.7187	0.1464	0.4643	-0.0311
2026-10-20	0.7355	0.1521	0.4813	-0.0352
2026-10-21	0.7272	0.1517	0.4847	-0.0352
2026-10-22	0.6962	0.1597	0.3497	-0.0650
2026-10-23	0.6937	0.1596	0.3433	-0.0658
2026-10-26	0.6986	0.1559	0.3376	-0.0619

2026-10-27	0.7026	0.1553	0.3221	-0.0616
2026-10-28	0.7139	0.1556	0.3657	-0.0614
2026-10-29	0.7193	0.1571	0.3430	-0.0679
2026-10-30	0.7193	0.1561	0.3275	-0.0620